CHERN CLASSES OF A FINITE LOCALLY FREE SHEAF ARE THE UNIQUE BIVARIANT CLASSES SATISFYING THE PROJECTIVE BUNDLE RELATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a uniqueness argument from the projective bundle formula followed by an inductive construction of the Chern classes out of the bivariant Segre classes

Editorial note. The label lemma-segre-classes names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The projective bundle formula, describing the Chow groups of a projective space bundle as a free module over those of the base, and the computation of the pushforward of powers of the first Chern class of the tautological quotient
- The notion of a bivariant class, and the closure of bivariance under capping with a first Chern class, under flat pullback and under proper pushforward
- The Segre class construction assigning to every base change of the projective bundle a map by pushing forward a power of the Chern class capped with the pullback

Source reference. Original reference 0B77: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 4144–4144. Prerequisite pointer invoked at proof line(s) 4633; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TW: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 4321–4321. Prerequisite pointer invoked at proof line(s) 4636; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TX: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 4381–4381. Prerequisite pointer invoked at proof line(s) 4622; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 0B78: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 4160–4160. Prerequisite pointer invoked at proof line(s) 4634; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EPK: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 4194–4194. Prerequisite pointer invoked at proof line(s) 4635; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 4598; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Let $\mathcal{E}$ be a finite locally free sheaf of rank r on X. Let $(\pi : P \to X, \mathcal{O}_P(1))$ be the projective space bundle associated to $\mathcal{E}$. For every morphism $X' \to X$ of good algebraic spaces over B there are unique maps*

$$c_i(\mathcal{E}) \cap - : \mathrm{CH}_k(X') \longrightarrow \mathrm{CH}_{k-i}(X'), \quad i = 0, \ldots, r$$

such that for $\alpha \in \mathrm{CH}_k(X')$ we have $c_0(\mathcal{E}) \cap \alpha = \alpha$ and

$$\sum\nolimits_{i=0,\ldots,r} (-1)^i c_1(\mathcal{O}_{P'}(1))^i \cap (\pi')^* \left(c_{r-i}(\mathcal{E}) \cap \alpha\right) = 0$$

where $\pi' : P' \to X'$ is the base change of π. Moreover, these maps define a bivariant class $c_i(\mathcal{E})$ of degree i on X.

Proof. Uniqueness and existence of the maps $c_i(\mathcal{E}) \cap -$ follows immediately from Lemma 02TX and the given description of $c_0(\mathcal{E})$. For every $i \in \mathbf{Z}$ the rule which to every morphism $X' \to X$ of good algebraic spaces over B assigns the map

$$t_i(\mathcal{E}) \cap - : \mathrm{CH}_k(X') \longrightarrow \mathrm{CH}_{k-i}(X'), \quad \alpha \longmapsto \pi'_*(c_1(\mathcal{O}_{P'}(1))^{r-1+i} \cap (\pi')^*\alpha)$$

is a bivariant class[1] by Lemmas 0B77, 0B78, and 0EPK. By Lemma 02TW we have $t_i(\mathcal{E}) = 0$ for $i < 0$ and $t_0(\mathcal{E}) = 1$. Applying pushforward to the equation in the statement of the lemma we find from Lemma 02TW that

$$(-1)^r t_1(\mathcal{E}) + (-1)^{r-1} c_1(\mathcal{E}) = 0$$

In particular we find that $c_1(\mathcal{E})$ is a bivariant class. If we multiply the equation in the statement of the lemma by $c_1(\mathcal{O}_{P'}(1))$ and push the result forward to X' we find

$$(-1)^r t_2(\mathcal{E}) + (-1)^{r-1} t_1(\mathcal{E}) \cap c_1(\mathcal{E}) + (-1)^{r-2} c_2(\mathcal{E}) = 0$$

As before we conclude that $c_2(\mathcal{E})$ is a bivariant class. And so on. □

[1] Up to signs these are the Segre classes of $\mathcal{E}$.